

B.TECH.- THIRD SEMESTER
MID-SEMESTER EXAMINATION, SEPTEMBER 2025

Course Code-ICMTC301/EEMTC301/CMMTC301
 Course Title-Engineering Mathematics

Time: 1:30 hours

Max Marks-20

Note: Attempt all questions in given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO																
1(a)	In the Fourier series expansion of the function $f(x) = e^{ax} = \frac{a_0}{2} + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$ on the interval $(0, 2\pi)$, find a_0 and a_n .	2	CO1																
1(a)	Find the half range sine series of the function $f(x) = \frac{\pi-x}{2}$ in the interval $(0, \pi)$.	2	CO1																
2(a)	Find the Harmonics a_0 and a_1 by using the following data: <table border="1"><tr><td>x</td><td>0</td><td>$\frac{\pi}{3}$</td><td>$\frac{2\pi}{3}$</td><td>π</td><td>$\frac{4\pi}{3}$</td><td>$\frac{5\pi}{3}$</td><td>2π</td></tr><tr><td>y</td><td>1</td><td>1.4</td><td>1.9</td><td>1.7</td><td>1.5</td><td>1.2</td><td>1</td></tr></table>	x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π	y	1	1.4	1.9	1.7	1.5	1.2	1	2	CO1
x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π												
y	1	1.4	1.9	1.7	1.5	1.2	1												
2(b)	Find the Fourier series of $f(x) = \pi x$ in the interval $(-c, c)$.	2	CO1																
3(a)	Determine the constant a such that the vector $\vec{A} = (ax^2y + yz)\hat{i} + (xy^2 - xz^2)\hat{j} + (2xyz - 2x^2y^2)\hat{k}$ is solenoidal.	2	CO2																
3(b)	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = \vec{r} $, then show that $\text{grad} \vec{r} ^3 = 3r\vec{r}$.	2	CO2																
4(a)	For the function $f(x, y) = \frac{y}{x^2+y^2}$, find the value of the directional derivative making an angle 30° with the positive x -axis at the point $(0, 1)$.	2	CO2																

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4(b)	Verify Green's theorem for the integral $\int_C (x + 2y)dx + (y + 3x)dy$, where C is the boundary of the region surrounded by the circle $x^2 + y^2 = 1$.	2	CO2
5(a)	Find the work done in moving a particle once around the circle $x^2 + y^2 = 9$ in the xy -plane if the field of force is $\vec{F} = (2x - y - z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y + 4z)\hat{k}$.	2	CO2
5(b)	Evaluate $\iint_S \vec{r} \cdot \hat{n} dS$, where S is the surface of the cube bounded by $x = -1, y = -1, z = -1, x = 1, y = 1$ and $z = 1$.	2	CO2
